# Chapter 4. Brownian Motion

Definition and path properties, Lévy's construction, the Markov property, $d$-dimensional Brownian motion

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## 4.1 Definition and some properties

**Motivation.** The continuous limit of the random walk (Example 3.8): quadratic variation $t$, hence infinite length.

For $I \subset \mathbb{R}_+ := [0,\infty)$ (e.g. $\mathbb{N}_0$, $[0,T]$, dyadic rationals $\mathbb{Q}_2^+ := \{m2^{-n} \mid m,n \geq 0\}$, $\mathbb{R}_+$), a **stochastic process** is a family $(X_t)_{t\in I}$ of r.v.'s on $(\Omega,\mathcal{F},P)$, possibly with values in a metric **state space** $S$; $t \mapsto X_t(\omega)$ is a **sample path**.

**Definition 4.1 (Brownian motion)**

A real-valued process $(B_t)_{t\geq 0}$ is a (standard) **Brownian motion** (BM) if

(i) $P(B_0 = 0) = 1$;

(ii) for $0 \leq s < t$, $B_t - B_s \sim N(0, t-s)$, i.e. $P(B_t - B_s \leq b) = \frac{1}{\sqrt{2\pi(t-s)}} \int_{-\infty}^{b} e^{-x^2/(2(t-s))} dx$;

(iii) it has **independent increments**: for $0 \leq t_1 < \cdots < t_n$, the r.v.'s $B_{t_1} - B_0, B_{t_2} - B_{t_1}, \ldots, B_{t_n} - B_{t_{n-1}}$ are independent;

(iv) almost all paths are continuous: $P(t \mapsto B_t \text{ is continuous on } [0,\infty)) = 1$.

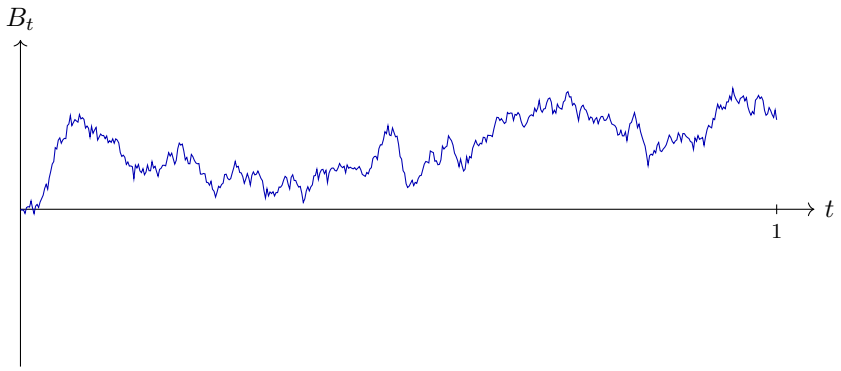



**Figure.** *A simulated Brownian path on $[0,1]$ (2000 steps); quadratic variation $t$ (Theorem 4.3), infinite total variation (Proposition 4.6).*

**Proposition 4.2**

Let $(B_t)_{t\geq 0}$ be a BM. (i) (Translation invariance) For fixed $t_0 \geq 0$, $\tilde{B}_t := B_{t+t_0} - B_{t_0}$ is a BM. (ii) (Scaling invariance) For $\lambda > 0$, $\tilde{B}_t := \frac{1}{\sqrt{\lambda}} B_{\lambda t}$ is a BM.

**Proof.** (i) Definition 4.1(i),(iv) are clear, and $\tilde{B}_t - \tilde{B}_s = B_{t+t_0} - B_{s+t_0} \sim N(0, t-s)$. For $0 \leq t_1 < \cdots < t_n$, the increments of $\tilde{B}$ are increments of $B$ along $t_0 \leq t_1 + t_0 < \cdots < t_n + t_0$ (if $t_1 = 0$ the first one is the constant 0, which is independent of everything), hence independent by (iii) for $B$.

(ii) (i), (iii),

(iv) carry over directly. For (ii), substituting $y = x/\sqrt{\lambda}$,

$$P(\tilde{B}_t - \tilde{B}_s \leq b) = P(B_{\lambda t} - B_{\lambda s} \leq \sqrt{\lambda} b) = \int_{-\infty}^{\sqrt{\lambda} b} \frac{e^{-x^2/(2\lambda(t-s))}}{\sqrt{2\pi\lambda(t-s)}} dx = \int_{-\infty}^{b} \frac{e^{-y^2/(2(t-s))}}{\sqrt{2\pi(t-s)}} dy.$$

■

Let $\tau_n = \{0 = t_1^{(n)} < t_2^{(n)} < \cdots < t_{N(\tau_n)}^{(n)} < \infty\}$ be partitions of $[0,\infty)$ with

$$|\tau_n| := \max_i \left(t_{i+1}^{(n)} - t_i^{(n)}\right) \to 0 \quad \text{(mesh)}, \qquad t_{N(\tau_n)}^{(n)} \to \infty. \tag{$\star$}$$

**Theorem 4.3 (Quadratic variation)**

Let $(B_t)$ be a BM and $(\tau_n)$ as in $(\star)$.

(i) For every $t \geq 0$, $S_t^n := \sum_{t_i^{(n)} \in \tau_n,\ t_{i+1}^{(n)} \leq t} \left(B_{t_{i+1}^{(n)}} - B_{t_i^{(n)}}\right)^2 \to t$ in $L^2(\Omega, \mathcal{F}, P)$.

(ii) If $\tau_n \subset \tau_{n+1}$ for all $n$, or $\sum_n |\tau_n| < \infty$, then $P(\lim_n S_t^n = t\ \forall t \geq 0) = 1$.

**Proof.** (i) Write $\Delta_i := t_{i+1}^{(n)} - t_i^{(n)}$; the sums run over $i$ with $t_{i+1}^{(n)} \leq t$. Then $\mathbb{E}[S_t^n] = \sum \Delta_i \to t$ in $\mathbb{R}$, by $(\star)$. By independence of increments and $\mathbb{E}[N^4] = 3\sigma^4$ for $N \sim N(0, \sigma^2)$,

$$\operatorname{Var}(S_t^n) = \sum \operatorname{Var}\left((B_{t_{i+1}} - B_{t_i})^2\right) = \sum (3\Delta_i^2 - \Delta_i^2) = 2\sum \Delta_i^2 \leq 2t|\tau_n| \to 0.$$

Hence $S_t^n - \mathbb{E}[S_t^n] \to 0$ in $L^2$, and with $\mathbb{E}[S_t^n] \to t$ we get $S_t^n \to t$ in $L^2$.

(ii) [Supplement] *Case* $\sum_n |\tau_n| < \infty$ (this includes the dyadic partitions $t_i^{(n)} = (i-1)2^{-n}$, the case treated in the lectures).

*Step 1 (fixed $t$).* By Chebyshev and the bound above, $\sum_n P(|S_t^n - \mathbb{E}S_t^n| \geq \varepsilon) \leq \varepsilon^{-2} \sum_n 2t|\tau_n| < \infty$. By Lemma 2.10, $S_t^n - \mathbb{E}[S_t^n] \to 0$ a.s., hence $S_t^n \to t$ on a set $\Omega_t$ with $P(\Omega_t) = 1$.

*Step 2 (all $t$ at once).* Let $\Omega' := \bigcap_{t \in \mathbb{Q}_2^+} \Omega_t$, so $P(\Omega') = 1$. For $\omega \in \Omega'$ and arbitrary $t$, take dyadic $t_k^- \nearrow t$, $t_k^+ \searrow t$. Since $S_t^n$ is increasing in $t$,

$$t_k^- = \lim_n S_{t_k^-}^n(\omega) \leq \liminf_n S_t^n(\omega) \leq \limsup_n S_t^n(\omega) \leq \lim_n S_{t_k^+}^n(\omega) = t_k^+,$$

and $k \to \infty$ gives the claim.

*General nested partitions*: see Revuz–Yor, *Continuous Martingales and Brownian Motion*, Ch. II, Prop. (2.12); not proved in the lectures. ∎

**Remark 4.4**

Let $\Omega''$ with $P(\Omega'') = 1$ be such that $t \mapsto B_t(\omega)$ is continuous for $\omega \in \Omega''$, and $\tilde{\Omega} := \Omega' \cap \Omega''$ with $\Omega'$ from the proof above. For $\tilde{\omega} \in \tilde{\Omega}$ the path is continuous and, along $(\tau_n)$ as in Theorem 4.3(ii),

$$\langle B \rangle_t(\tilde{\omega}) := \lim_n \sum_{t_{i+1}^{(n)} \leq t} \left(B_{t_{i+1}^{(n)}}(\tilde{\omega}) - B_{t_i^{(n)}}(\tilde{\omega})\right)^2 = t \qquad \forall t \geq 0.$$

Such a path is a **typical path**, and $\langle B \rangle_t(\tilde{\omega})$ its **quadratic variation**.

**Definition 4.5 (Finite variation)**

$g : \mathbb{R}_+ \to \mathbb{R}$ has **finite variation** on $[0, t]$ if

$$\operatorname{Var}_t g := \sup_{\tau} \sum_{t_i \in \tau,\ t_{i+1} \leq t} |g(t_{i+1}) - g(t_i)| < \infty, \tag{4.1}$$

the supremum over all partitions $0 = t_1 < \cdots < t_{N(\tau)} < \infty$; $g$ has finite variation if this holds for every $t$.

**Proposition 4.6**

A typical path of BM is continuous and of unbounded variation on every interval $[0, T]$, $T > 0$.

**Proof.** Suppose $\mathrm{Var}_T B_\cdot(\tilde\omega) < \infty$ for some $T > 0$. The path is uniformly continuous on $[0,T]$, so for $\varepsilon > 0$ the increments are $< \varepsilon$ once $|\tau_n| < \delta$, and

$$\sum_{t_{i+1}\le T} (B_{t_{i+1}} - B_{t_i})^2(\tilde\omega) \le \max_i |B_{t_{i+1}} - B_{t_i}|(\tilde\omega) \cdot \sum_i |B_{t_{i+1}} - B_{t_i}|(\tilde\omega) \le \varepsilon \,\mathrm{Var}_T B_\cdot(\tilde\omega).$$

Hence $\langle B\rangle_T(\tilde\omega) = 0$, contradicting $\langle B\rangle_T(\tilde\omega) = T > 0$. ∎

## 4.2 Construction (hence existence) of Brownian motion

**Motivation.** Build the process on dyadic times (Kolmogorov extension), prove continuity (Kolmogorov–Čentsov), and extend.

For a process $(X_t)_{t\in I}$ and pairwise distinct $t_1,\dots,t_n \in I$ let

$$\mu_{t_1\dots t_n}(A_1\times\cdots\times A_n) := P((X_{t_1},\dots,X_{t_n}) \in A_1\times\cdots\times A_n), \qquad A_i \in \mathcal{B}(\mathbb{R}). \tag{4.2}$$

By Theorem 1.9 this determines the law $P((X_{t_1},\dots,X_{t_n}) \in A)$, $A \in \mathcal{B}(\mathbb{R}^n)$, because rectangles are a $\cap$-stable generator of $\mathcal{B}(\mathbb{R}^n)$. These **finite-dimensional distributions** are **consistent**:

$$\mu_{t_1\dots t_n}(A_1\times\cdots\times A_n) = \mu_{t_{\pi(1)}\dots t_{\pi(n)}}(A_{\pi(1)}\times\cdots\times A_{\pi(n)}) \quad \text{for permutations } \pi, \tag{4.3}$$

$$\mu_{t_1\dots t_n}(A_1\times\cdots\times A_{n-1}\times\mathbb{R}) = \mu_{t_1\dots t_{n-1}}(A_1\times\cdots\times A_{n-1}) \quad (n\ge 2). \tag{4.4}$$

Any family of probability measures $\mu_{t_1\dots t_n}$ on $\mathbb{R}^n$ satisfying (4.3), (4.4) is called consistent. Conversely, does every consistent family come from a process?

On $\mathbb{R}^I := \{\omega : I \to \mathbb{R}\}$ let the **cylinder sets** be

$$\mathcal{C}_I := \{\{\omega \mid (\omega(t_1),\dots,\omega(t_n)) \in A_1\times\cdots\times A_n\} \mid t_1 < \cdots < t_n \in I,\ A_i \in \mathcal{B}(\mathbb{R}),\ n\in\mathbb{N}\}.$$

$\sigma(\mathcal{C}_I)$ is the smallest $\sigma$-algebra making all finite-dimensional projections $f_{t_1\dots t_n}(\omega) := (\omega(t_1),\dots,\omega(t_n))$ measurable, and, since $f^{-1}_{t_1\dots t_n}(A_1\times\cdots\times A_n) = \bigcap_i f_{t_i}^{-1}(A_i)$, even the smallest one making all $f_t$, $t\in I$, measurable.

**Theorem 4.7 (Kolmogorov's extension theorem)**

For every consistent family $\mu_{t_1\dots t_n}$ there is a unique probability measure $P$ on $(\mathbb{R}^I, \sigma(\mathcal{C}_I))$ such that, with $X_t(\omega) := \omega(t)$, (4.2) holds, i.e. $\mu_{t_1\dots t_n} = P \circ f^{-1}_{t_1\dots t_n}$.

**Proof [Supplement].**

*Step 1 (inner regularity).* Every probability measure $\mu$ on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ satisfies $\mu(A) = \sup\{\mu(K) \mid K \subset A \text{ compact}\}$. The class of $A$ such that for every $\varepsilon$ there are $F$ closed $\subset A \subset G$ open with $\mu(G\setminus F) < \varepsilon$ is a $\sigma$-algebra. It contains every closed $F$ (take $G = \{d(\cdot,F) < \frac{1}{m}\}$ and continuity from above), hence all Borel sets. Intersecting $F$ with a large closed ball gives a compact $K$.

*Step 2 (a finitely additive $P_0$).* The general cylinders $Z = \{\omega \mid f_J(\omega) \in A\}$, $J \subset I$ finite, $A \in \mathcal{B}(\mathbb{R}^J)$, form an algebra $\mathcal{Z}$ with $\sigma(\mathcal{Z}) = \sigma(\mathcal{C}_I)$. Put $P_0(Z) := \mu_J(A)$. This is well defined: two representations can be lifted to a common finite $J' \supset J$, and (4.3), (4.4), which extend from rectangles to all Borel sets by Theorem 1.9, show that $\mu_J(A) = \mu_{J'}(A\times\mathbb{R}^{J'\setminus J})$. Since finitely many cylinders can be written over one common $J$, $P_0$ is finitely additive with $P_0(\mathbb{R}^I) = 1$.

*Step 3 ($\sigma$-additivity).* By Theorem 1.15 it suffices that $Z_k \in \mathcal{Z}$, $Z_k \searrow \emptyset$ imply $P_0(Z_k) \to 0$. Suppose $P_0(Z_k) \ge \varepsilon > 0$ for all $k$, with $Z_k = \{f_{J_k} \in A_k\}$ and $J_1 \subset J_2 \subset \cdots$. By Step 1 choose compact $K_k \subset A_k$ with $\mu_{J_k}(A_k \setminus K_k) \le \varepsilon 2^{-k-1}$, and put $E_k := \bigcap_{j\le k}\{f_{J_j} \in K_j\} \subset Z_k$. Then $P_0(Z_k\setminus E_k) \le \varepsilon/2$, so $P_0(E_k) \ge \varepsilon/2$ and $E_k \ne \emptyset$; pick $\omega_k \in E_k$. On the countable set $J := \bigcup_k J_k$,

for each $t \in J_j$ the values $\omega_k(t)$, $k \geq j$, lie in the compact projection of $K_j$. A diagonal argument gives a subsequence with $\omega_{k_l}(t) \to \omega(t)$ for all $t \in J$; put $\omega(t) := 0$ off $J$. For each $j$, $f_{J_j}(\omega_{k_l}) \in K_j$ for $k_l \geq j$ and $K_j$ is closed, so $f_{J_j}(\omega) \in K_j \subset A_j$. Thus $\omega \in \bigcap_j Z_j = \emptyset$, a contradiction.

*Step 4.* Carathéodory's Theorem 1.11 (on the algebra $\mathcal{Z}$, finite measure) extends $P_0$ uniquely to $\sigma(\mathcal{C}_I)$, and $P \circ f_{t_1 \dots t_n}^{-1} = \mu_{t_1 \dots t_n}$ by construction. Uniqueness: $\mathcal{Z}$ is a $\cap$-stable generator (Theorem 1.9). ∎

**The Brownian finite-dimensional laws.** Let $I = \mathbb{R}_+$ and

$$p_t(x,y) := \frac{1}{\sqrt{2\pi t}} e^{-\frac{(y-x)^2}{2t}} \quad (t>0), \qquad p_0(x,y)\,dy := \delta_x(dy). \tag{4.5}$$

For $0 \leq t_1 < \cdots < t_n$ and $A \in \mathcal{B}(\mathbb{R}^n)$ set, with $t_0 := 0$, $x_0 := 0$,

$$\mu_{t_1 \dots t_n}(A) := \int \cdots \int \mathbb{1}_A(x_1, \dots, x_n) \prod_{i=1}^{n} p_{t_i - t_{i-1}}(x_{i-1}, x_i)\, dx_n \cdots dx_1$$

(if $t_1 = 0$, the first factor means $\delta_0(dx_1)$). This is a probability measure: the integrand is a nonnegative Borel density (Theorem 1.47), and integrating out $x_n, x_{n-1}, \dots$ successively gives total mass 1. By Fubini and Theorem 1.9 it is the unique probability measure with

$$\mu_{t_1 \dots t_n}(A_1 \times \cdots \times A_n) = \int_{A_1} \cdots \int_{A_n} \prod_{i=1}^{n} p_{t_i - t_{i-1}}(x_{i-1}, x_i)\, dx_n \cdots dx_1. \tag{4.6}$$

(Replacing $x_0 = 0$ by $x_0 = x$ gives Brownian motion started at $x$.) For unordered times, sort them by the unique permutation $\sigma$ with $t_{\sigma(1)} < \cdots < t_{\sigma(n)}$ and permute the coordinates accordingly. The family is consistent: (4.3) holds by construction, and (4.4) amounts to deleting one time,

$$\mu_{t_1 \dots t_{i-1} t_{i+1} \dots t_n}(\cdots \times A_{i-1} \times A_{i+1} \times \dots) = \mu_{t_1 \dots t_n}(\cdots \times A_{i-1} \times \mathbb{R} \times A_{i+1} \times \dots). \tag{4.7}$$

For $i = n$ this is immediate. For $i < n$ it is the **Chapman–Kolmogorov equation**

$$\int_A p_{t+s}(x,z)\, dz = \int_A \int_{\mathbb{R}} p_s(x,y) p_t(y,z)\, dy\, dz, \qquad \text{i.e.} \qquad \int p_s(x,y) p_t(y,z)\, dy = p_{s+t}(x,z),$$

which holds because $N(0,s) * N(0,t) = N(0, s+t)$ (Example 3.44(i)). [Supplement]

**The difficulty.** Theorem 4.7 gives $\tilde{P}$ on $(\mathbb{R}^{\mathbb{R}_+}, \sigma(\mathcal{C}_{\mathbb{R}_+}))$ with (i)–(iii) of Definition 4.1, but $C(\mathbb{R}_+) \notin \sigma(\mathcal{C}_{\mathbb{R}_+})$ (its sets depend on countably many coordinates). So restrict to $\mathbb{Q}_2^+$: the family is consistent and Theorem 4.7 gives a unique $Q$ on $(\mathbb{R}^{\mathbb{Q}_2^+}, \sigma(\mathcal{C}_{\mathbb{Q}_2^+}))$.

**Theorem 4.8**

Let $C(\mathbb{Q}_2^+) := \bigcap_{n \geq 1} \{\omega \in \mathbb{R}^{\mathbb{Q}_2^+} \mid \omega|_{[0,n] \cap \mathbb{Q}_2^+} \text{ is uniformly continuous}\}$. Then $C(\mathbb{Q}_2^+) \in \sigma(\mathcal{C}_{\mathbb{Q}_2^+})$ and $Q(C(\mathbb{Q}_2^+)) = 1$; so $\big(C(\mathbb{Q}_2^+), C(\mathbb{Q}_2^+) \cap \sigma(\mathcal{C}_{\mathbb{Q}_2^+}), Q\big)$ is a probability space (with the trace $\sigma$-algebra).

**From Theorem 4.8 to Brownian motion.** Let $\Omega := C(\mathbb{R}_+)$ with the metric

$$\rho(\omega, \omega') := \sum_{n \geq 1} 2^{-n} \Big(1 \wedge \sup_{t \in [0,n]} |\omega(t) - \omega'(t)|\Big),$$

under which $\Omega$ is a complete separable metric space; $\mathcal{F} := \mathcal{B}(\Omega)$ and $B_t(\omega) := \omega(t)$, which is $\rho$-continuous and hence measurable, so $\sigma(B_t \mid t \geq 0) \subset \mathcal{F}$. Conversely, for a countable dense $D \subset \Omega$ and $\omega_0 \in D$, $\rho(\omega, \omega_0) = \sum_n 2^{-n}(1 \wedge \sup_{t \in [0,n] \cap \mathbb{Q}_2^+} |B_t(\omega) - \omega_0(t)|)$ is $\sigma(B_t \mid t \in \mathbb{Q}_2^+)$-measurable. Every open set is a countable union of balls $B_\rho(\omega_0, r)$, $\omega_0 \in D$, $r \in \mathbb{Q}_2^+$, so $\mathcal{F} = \sigma(B_t \mid t \geq 0)$.

The map $\psi : C(\mathbb{Q}_2^+) \to \Omega$, $\omega \mapsto$ the unique continuous extension of $\omega$ to $\mathbb{R}_+$, is measurable: for $q_m \to t$ in $\mathbb{Q}_2^+$,

$$\psi^{-1}(\{B_t < \alpha\}) = \bigcup_k \bigcup_n \bigcap_{m \ge n} \{\omega \mid \omega(q_m) < \alpha - \tfrac{1}{k}\},$$

and the sets $\{B_t < \alpha\}$ generate $\mathcal{F}$. Let $P := Q \circ \psi^{-1}$. For dyadic $t_i$, $(\psi\omega)(t_i) = \omega(t_i)$, so

$$P((B_{t_1}, \dots, B_{t_n}) \in A_1 \times \cdots \times A_n) = Q((X_{t_1}, \dots, X_{t_n}) \in A_1 \times \cdots \times A_n) \qquad (t_i \in \mathbb{Q}_2^+). \tag{4.8}$$

Then $B_t(\omega) = \omega(t)$ is a Brownian motion on $(\Omega, \mathcal{F}, P)$, every path being continuous. The key is that for all $0 \le t_1 < \cdots < t_n$ in $\mathbb{R}_+$, with $t_0 = 0$, $x_0 = 0$,

$$P((B_{t_1}, \dots, B_{t_n}) \in A_1 \times \cdots \times A_n) = \int_{A_1} \cdots \int_{A_n} \prod_{i=1}^{n} p_{t_i - t_{i-1}}(x_{i-1}, x_i)\, dx_n \cdots dx_1, \tag{4.9}$$

and that (4.9) is equivalent to Definition 4.1(i)–(iii).

[Supplement] *Proof of* (4.9). Take dyadic $t_i^k \to t_i$ (strictly ordered). For $f \in C_b(\mathbb{R}^n)$, $\mathbb{E}_P[f(B_{t_1^k}, \dots, B_{t_n^k})] = \int f \, d\mu_{t_1^k \dots t_n^k}$ by (4.8). As $k \to \infty$ the left side tends to $\mathbb{E}_P[f(B_{t_1}, \dots, B_{t_n})]$ by path continuity and the DCT. In the right side substitute $x_i = x_{i-1} + \sqrt{t_i^k - t_{i-1}^k}\, z_i$: it becomes an integral of $f$ evaluated at points depending continuously on $k$ against a fixed Gaussian density, so the DCT gives the limit $\int f \, d\mu_{t_1 \dots t_n}$. By Example 2.30 on $\mathbb{R}^n$ the two laws agree.

(4.9) *gives (i)–(iii).* $n = 1$ gives $B_t \sim N(0, t)$ and $B_0 = 0$ a.s. For $t_1 < t_2$, with $\bar{A} = \{(x, y) \mid y - x \le \alpha\}$ and $z = y - x$,

$$P(B_{t_2} - B_{t_1} \le \alpha) = \int_{\mathbb{R}} p_{t_1}(0, x) \int_{-\infty}^{x+\alpha} p_{t_2 - t_1}(x, y)\, dy\, dx = \int_{-\infty}^{\alpha} \frac{e^{-z^2/(2(t_2 - t_1))}}{\sqrt{2\pi(t_2 - t_1)}} dz.$$

More generally, the change of variables $z_i = x_i - x_{i-1}$ turns the density in (4.9) into $\prod_i p_{t_i - t_{i-1}}(0, z_i)$, a product density. So $(B_{t_1}, B_{t_2} - B_{t_1}, \dots, B_{t_n} - B_{t_{n-1}})$ has independent $N(0, t_i - t_{i-1})$ coordinates (Theorem 3.42), i.e.

$$P(B_{t_1} \le \alpha_1, B_{t_2} - B_{t_1} \le \alpha_2, \dots, B_{t_n} - B_{t_{n-1}} \le \alpha_n) = P(B_{t_1} \le \alpha_1) \cdots P(B_{t_n} - B_{t_{n-1}} \le \alpha_n). \tag{4.10}$$

Conversely (i)–(iii) determine the joint law of the increments, hence (4.9).

**Proof of Theorem 4.8.**

*Step 1 (measurability).* With $\Omega_n := \{\omega \mid \omega|_{[0,n] \cap \mathbb{Q}_2^+}$ uniformly continuous$\}$,

$$\Omega_n = \bigcap_{M \ge 1} \bigcup_{N \ge 1} \bigcap_{t,s \in \mathbb{Q}_2^+ \cap [0,n],\ |t-s| \le \frac{1}{N}} \left\{ |X_t - X_s| \le \frac{1}{M} \right\} \in \sigma(\mathcal{C}_{\mathbb{Q}_2^+}),$$

a countable combination of cylinder sets. It remains to show $Q(\Omega_n) = 1$ for each $n$.

*Step 2 (moments).* The marginals of $(X_t)_{t \in \mathbb{Q}_2^+}$ satisfy Definition 4.1(i)–(iii), hence scaling invariance: $X_t - X_s \sim \frac{1}{\sqrt{\lambda}}(X_{\lambda t} - X_{\lambda s}) \sim N(0, t - s)$. For $Y_t^n := \frac{1}{\sqrt{n}} X_{nt}$ and $s < t$ in $\mathbb{Q}_2^+ \cap [0, 1]$,

$$\mathbb{E}_Q |Y_t^n - Y_s^n|^{2k} = \mathbb{E}_Q |X_t - X_s|^{2k} = \mathbb{E}_Q |\sqrt{t - s}\, X_1|^{2k} = c_k |t - s|^k, \qquad c_k := \mathbb{E}_Q |X_1|^{2k}. \tag{4.11}$$

*Step 3.* By Theorem 4.9 and Remark 4.10 there is $\Omega^*$ with $Q(\Omega^*) = 1$ on which $t \mapsto Y_t^n(\omega)$ is Hölder continuous of order $\gamma \in (0, \frac{1}{2})$ on $[0, 1] \cap \mathbb{Q}_2^+$, i.e. $t \mapsto X_t(\omega)$ is Hölder on $[0, n] \cap \mathbb{Q}_2^+$. So $\Omega^* \subset \Omega_n$. ∎

**Theorem 4.9 (Kolmogorov–Čentsov, dyadic version)**

Let $(X_t)_{t\in\mathbb{Q}_2^+}$ be a process on $(\mathbb{R}^{\mathbb{Q}_2^+}, \sigma(\mathcal{C}_{\mathbb{Q}_2^+}), Q)$ with

$$\mathbb{E}_Q|X_t - X_s|^\alpha \le K|t-s|^{1+\beta} \qquad \forall t,s\in\mathbb{Q}_2^+\cap[0,1]$$

for some $\alpha,\beta,K>0$, and let $\gamma\in(0,\beta/\alpha)$. Then there is $\Omega^*\in\sigma(\mathcal{C}_{\mathbb{Q}_2^+})$ with $Q(\Omega^*)=1$ such that for every $\omega\in\Omega^*$ there is $n^*(\omega)$ with

$$\sup_{t,s\in\mathbb{Q}_2^+\cap[0,1],\ 0<t-s<2^{-n^*(\omega)}} \frac{|X_t(\omega)-X_s(\omega)|}{|t-s|^\gamma} \le C := \frac{2}{1-2^{-\gamma}}.$$

In particular every $\omega\in\Omega^*$ is Hölder continuous of order $\gamma$ on $\mathbb{Q}_2^+\cap[0,1]$: with $n_0 := n^*(\omega)+1$, splitting $[s,t]$ into $2^{n_0}$ equal pieces gives $|X_t(\omega)-X_s(\omega)| \le 2^{n_0}C\big((t-s)2^{-n_0}\big)^\gamma = 2^{n_0(1-\gamma)}C|t-s|^\gamma$.

**Remark 4.10**

By (4.11) we may take $\alpha=2k$, $\beta=k-1$, and $\beta/\alpha=\frac{k-1}{2k}\nearrow\frac12$. So Theorem 4.9 applies to every $\gamma\in(0,\frac12)$: $Q$-a.e. path is locally Hölder continuous of every order $\gamma<\frac12$. The map $\psi$ preserves local Hölder continuity, so $P$-a.e. Brownian path in $C(\mathbb{R}_+)$ is locally $\gamma$-Hölder for every $\gamma<\frac12$.

**Proof of Theorem 4.9.**

*Step 1 (Borel–Cantelli on dyadic increments).* By Markov's inequality and the hypothesis,

$$\begin{aligned} Q\Big(\max_{1\le k\le 2^n}|X_{k2^{-n}}-X_{(k-1)2^{-n}}|\ge 2^{-\gamma n}\Big) &\le \sum_{k=1}^{2^n}\frac{\mathbb{E}_Q|X_{k2^{-n}}-X_{(k-1)2^{-n}}|^\alpha}{2^{-\gamma n\alpha}} \\ &\le 2^n\frac{K2^{-n(1+\beta)}}{2^{-\gamma n\alpha}} = K2^{-n(\beta-\gamma\alpha)}. \end{aligned}$$

This is summable since $\beta>\gamma\alpha$. By Borel–Cantelli, $\Omega^* := \bigcup_l\bigcap_{n\ge l}\{\max_k|X_{k2^{-n}}-X_{(k-1)2^{-n}}|<2^{-\gamma n}\}$ has $Q(\Omega^*)=1$, and for $\omega\in\Omega^*$ there is $n^*(\omega)$ with

$$\max_{1\le k\le 2^n}|X_{k2^{-n}}(\omega)-X_{(k-1)2^{-n}}(\omega)|<2^{-\gamma n} \qquad \forall n\ge n^*(\omega). \tag{4.12}$$

*Step 2 (chaining).* Fix $\omega\in\Omega^*$, $n\ge n^*(\omega)$, and let $D_m := \{k2^{-m}\mid 0\le k\le 2^m\}$. We claim, for all $m>n$,

$$|X_t(\omega)-X_s(\omega)| \le 2\sum_{j=n+1}^{m}2^{-\gamma j} \qquad \forall t,s\in D_m,\ 0<t-s<2^{-n}. \tag{4.13}$$

For $m=n+1$, $t,s$ are neighbours in $D_{n+1}$, and (4.12) gives $<2^{-\gamma(n+1)}$. Assume (4.13) for $m=M-1$, and let $s,t\in D_M$, $0<t-s<2^{-n}$. Put $s_1 := \min\{r\in D_{M-1}\mid r\ge s\}$ and $t_1 := \max\{r\in D_{M-1}\mid r\le t\}$. Then $s\le s_1\le t_1\le t$ and $s_1-s, t-t_1\in\{0,2^{-M}\}$, so (4.12) bounds $|X_{s_1}-X_s|$ and $|X_t-X_{t_1}|$ by $2^{-\gamma M}$, and the induction hypothesis bounds $|X_{t_1}-X_{s_1}|$. Summing gives (4.13) for $m=M$.

*Step 3.* For $t,s\in\mathbb{Q}_2^+\cap[0,1]$ with $0<t-s<2^{-n^*(\omega)}$ choose $m>n\ge n^*(\omega)$ with $t,s\in D_m$ and $2^{-(n+1)}\le t-s<2^{-n}$. By (4.13),

$$|X_t-X_s| \le 2\sum_{j\ge n+1}2^{-\gamma j} = \frac{2\cdot 2^{-\gamma(n+1)}}{1-2^{-\gamma}} \le \frac{2}{1-2^{-\gamma}}|t-s|^\gamma.$$

■

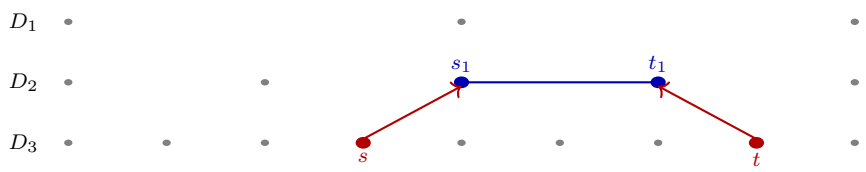



**Figure.** *Chaining (Theorem 4.9): one $2^{-j}$-step per level, increments $\le 2^{-\gamma j}$, a geometric series.*

**Remark 4.11**

$((B_t)_{t\ge0}, \Omega, \mathcal{F}, P)$ as constructed above is the **canonical model** of Brownian motion; $P = P_0$ is the **Wiener measure** (started at 0).

## 4.3 The Markov property

**Motivation.** Given $B_s$, the future is a Brownian motion from $B_s$; the filtration $\mathcal{F}_t^{0,+}$ adds Blumenthal's 0–1 law.

Work in the canonical model: $\Omega = C(\mathbb{R}_+)$, $B_t(\omega) = \omega(t)$, $\mathcal{F} = \mathcal{B}(C(\mathbb{R}_+)) = \sigma(B_t \mid t \ge 0)$. For a start in $x \in \mathbb{R}$ there are three equivalent options: (1) $(B_t)$ under $P_x$, built as $P$ but with $x_0 = x$ in (4.6); (2) $(B_t + x)$ under $P_0$; (3) $(B_t)$ under $P_0 \circ T_x^{-1}$ with $T_x(\omega) := \omega + x$. All have the same finite-dimensional distributions and continuous paths; in particular $P_x = P_0 \circ T_x^{-1}$. We write $\mathbb{E}_x[Y] := \int Y \, dP_x$.

Let $\mathcal{F}_t^0 := \sigma(B_u \mid u \le t)$. Then $(\mathcal{F}_t^0)_{t\ge0}$ is a **filtration** (information up to $t$) to which $(B_t)$ is **adapted**. The **shift** $\theta_s(\omega)(t) := \omega(s+t)$ is $\rho$-continuous and $\hat{\mathcal{F}}_s^0/\mathcal{F}$-measurable, $\hat{\mathcal{F}}_s^0 := \sigma(B_u \mid u \ge s)$; for $Y \in \mathcal{F}_b$, $Y \circ \theta_s$ is a function of the future after $s$, e.g. $Y = f(B_{t_1}, \dots, B_{t_n})$ gives $Y \circ \theta_s = f(B_{t_1+s}, \dots, B_{t_n+s})$, and $B_t \circ \theta_s = B_{t+s}$.

The canonical model has the **Markov property** w.r.t. $(\mathcal{F}_t^0)$: for $Y \in \mathcal{F}_b$,

$$\mathbb{E}_x[\underbrace{Y \circ \theta_s}_{\text{future}} \mid \underbrace{\mathcal{F}_s^0}_{\text{past}}] = \mathbb{E}_{\underbrace{B_s}_{\text{present}}}[Y] \qquad P_x\text{-a.s., } \forall s \ge 0,\ x \in \mathbb{R}. \tag{4.14}$$

It will follow from the Markov property (4.15) for a larger filtration (Remark 4.13).

**Remark 4.12**

$x \mapsto \mathbb{E}_x[Y]$ is Borel measurable. [Supplement] *Reason:* such $Y$ form an MVS (MCT) containing the cylinder indicators, whose $\mathbb{E}_x[\mathbb{1}_A]$ are continuous in $x$ ((4.6), DCT); apply Theorem 2.29. So $\mathbb{E}_{B_s}[Y]$ is $\sigma(B_s) \subset \mathcal{F}_s^0$-measurable, and $P_x$-a.s.

$$\begin{aligned}\mathbb{E}_x[Y \circ \theta_s \mid \sigma(B_s)] &= \mathbb{E}_x\big[\mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s^0] \,\big|\, \sigma(B_s)\big] = \mathbb{E}_x[\mathbb{E}_{B_s}[Y] \mid \sigma(B_s)] \\ &= \mathbb{E}_{B_s}[Y] = \mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s^0],\end{aligned}$$

using Theorem 2.35(iv), (4.14) and Example 2.34(i). The best guess for the future given the present equals the best guess given the whole past.

For technical reasons we enlarge $\mathcal{F}_t^0$ to

$$\mathcal{F}_t^{0,+} := \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^0 \ (\supset \mathcal{F}_t^0),$$

which allows an infinitesimal look into the future. It is **right continuous**: $\bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^{0,+} = \bigcap_n \bigcap_k \mathcal{F}_{t+\frac{1}{n}+\frac{1}{k}}^0 \subset \mathcal{F}_t^{0,+}$, and trivially $\mathcal{F}_t^{0,+} \subset \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^{0,+}$. $(B_t)$ is adapted to $(\mathcal{F}_t^{0,+})$ and has the

Markov property w.r.t. it:

$$\mathbb{E}_x[Y\circ\theta_s \mid \mathcal{F}_s^{0,+}] = \mathbb{E}_{B_s}[Y] \qquad P_x\text{-a.s., } \forall s\ge 0,\ x\in\mathbb{R},\ Y\in\mathcal{F}_b. \tag{4.15}$$

**Proof of (4.15).**

*Step 1 (reduction).* Let $\mathcal{C}$ be the cylinder sets $A=\{(B_{t_1},\dots,B_{t_n})\in A_1\times\cdots\times A_n\}$, $0<t_1<\cdots<t_n$, $A_i\in\mathcal{B}(\mathbb{R})$; $\mathcal{C}$ is $\cap$-stable and generates $\mathcal{F}$. $\mathcal{H}:=\{Y\in\mathcal{F}_b \mid (4.15)\text{ holds}\}$ is an MVS (conditional MCT), so by Theorem 2.29 it suffices to treat $Y=\mathbb{1}_A$, $A\in\mathcal{C}$, i.e. to show

$$\mathbb{E}_x[\mathbb{1}_A\circ\theta_s\,\mathbb{1}_C] = \mathbb{E}_x[\mathbb{E}_{B_s}[\mathbb{1}_A]\,\mathbb{1}_C] \qquad \forall C\in\mathcal{F}_s^{0,+}. \tag{4.16}$$

*Step 2 (special $C$).* Fix $x$, $s$ and $0<h<t_1$, and take $C=\{(B_{s_1},\dots,B_{s_m})\in C_1\times\cdots\times C_m\}$ with $0\le s_1<\cdots<s_m=s+h$. With $s_0:=0$, $y_0:=x$, (4.9) gives

$$\mathbb{E}_x[\mathbb{1}_A\circ\theta_s\,\mathbb{1}_C] = \int_{C_1}\cdots\int_{C_m}\prod_{j=1}^{m} p_{s_j-s_{j-1}}(y_{j-1},y_j)\,F(y_m,h)\,dy_m\cdots dy_1 = \mathbb{E}_x[F(B_{s+h},h)\mathbb{1}_C], \tag{4.17}$$

where, with $x_0:=y$ and $t_0:=h$,

$$F(y,h) := \int_{A_1}\cdots\int_{A_n} p_{t_1-h}(y,x_1)\prod_{i=2}^{n} p_{t_i-t_{i-1}}(x_{i-1},x_i)\,dx_n\cdots dx_1.$$

*Step 3 (all $C\in\mathcal{F}_{s+h}^0$).* The sets $C$ of Step 2 form a $\cap$-stable generator of $\mathcal{F}_{s+h}^0$, and the class of $\tilde C\in\mathcal{F}_{s+h}^0$ satisfying (4.17) is a Dynkin system. By Theorem 1.8,

$$\mathbb{E}_x[\mathbb{1}_A\circ\theta_s\,\mathbb{1}_C] = \mathbb{E}_x[F(B_{s+h},h)\mathbb{1}_C] \qquad \forall C\in\mathcal{F}_{s+h}^0\supset\mathcal{F}_s^{0,+}. \tag{4.18}$$

*Step 4 ($h\to 0$).* By the DCT, $F$ is jointly continuous in $(y,h)\in\mathbb{R}\times(0,t_1)$ and bounded by 1. As $h\to 0$, path continuity gives $F(B_{s+h}(\omega),h)\to\int_{A_1}\cdots\int_{A_n} p_{t_1}(B_s(\omega),x_1)\prod_{i\ge2}p_{t_i-t_{i-1}}(x_{i-1},x_i)\,dx = \mathbb{E}_{B_s(\omega)}[\mathbb{1}_A]$. The DCT in (4.18) then yields (4.16) for all $C\in\mathcal{F}_s^{0,+}$. ∎

**Remark 4.13**

(4.15) implies (4.14): $\mathbb{E}_x[Y\circ\theta_s\mid\mathcal{F}_s^0] = \mathbb{E}_x[\mathbb{E}_x[Y\circ\theta_s\mid\mathcal{F}_s^{0,+}]\mid\mathcal{F}_s^0] = \mathbb{E}_x[\mathbb{E}_{B_s}[Y]\mid\mathcal{F}_s^0] = \mathbb{E}_{B_s}[Y]$ by Theorem 2.35(iv), (4.15) and Example 2.34(i).

**Theorem 4.14**

For $Z\in\mathcal{F}_b$: $\mathbb{E}_x[Z\mid\mathcal{F}_s^{0,+}] = \mathbb{E}_x[Z\mid\mathcal{F}_s^0]$ $P_x$-a.s., for all $s\ge0$, $x\in\mathbb{R}$.

**Proof.** By the monotone class theorem it suffices to take $Z=\mathbb{1}_A$ with $A=\{(B_{t_1},\dots,B_{t_n})\in A_1\times\cdots\times A_n\}$, $0\le t_1<\cdots<t_n$. Then, $P_x$-a.s., by Theorem 2.35(iii) and (4.14), (4.15) (both equal $\mathbb{E}_{B_s}[\cdot]$),

$$\begin{aligned}\mathbb{E}_x[\mathbb{1}_A\mid\mathcal{F}_s^{0,+}] &= \prod_{t_i\le s}\mathbb{1}_{A_i}(B_{t_i})\,\mathbb{E}_x\Big[\prod_{t_i>s}\mathbb{1}_{A_i}(B_{t_i-s})\circ\theta_s\,\Big|\,\mathcal{F}_s^{0,+}\Big]\\ &= \prod_{t_i\le s}\mathbb{1}_{A_i}(B_{t_i})\,\mathbb{E}_x\Big[\prod_{t_i>s}\mathbb{1}_{A_i}(B_{t_i-s})\circ\theta_s\,\Big|\,\mathcal{F}_s^{0}\Big] = \mathbb{E}_x[\mathbb{1}_A\mid\mathcal{F}_s^0].\end{aligned}$$

∎

**Remark 4.15**

$\mathcal{F}_s^{0,+}$ and $\mathcal{F}_s^0$ differ only by $P_x$-null sets (for every $x$). For $Z \in \mathcal{F}_s^{0,+}$ bounded, $Z = \mathbb{E}_x[Z \mid \mathcal{F}_s^{0,+}] = \mathbb{E}_x[Z \mid \mathcal{F}_s^0]$ $P_x$-a.s., an $\mathcal{F}_s^0$-measurable r.v. With $Z = \mathbb{1}_{B^c}$, $B \in \mathcal{F}_s^{0,+}$, the set $B = \{\mathbb{1}_{B^c} < 1\}$ agrees up to a $P_x$-null set with $B_0 := \{\mathbb{E}_x[\mathbb{1}_{B^c} \mid \mathcal{F}_s^0] < 1\} \in \mathcal{F}_s^0$.

**Theorem 4.16 (Blumenthal's 0–1 law)**

If $A \in \mathcal{F}_0^{0,+}$, then $P_x(A) \in \{0,1\}$ for every $x \in \mathbb{R}$.

**Proof.** $\mathcal{F}_0^0 = \sigma(B_0)$ and $P_x(B_0 = x) = 1$, so every $\mathcal{F}_0^0$-measurable $Y$ is $P_x$-a.s. constant, $Y = \mathbb{E}_x[Y]$. For $A \in \mathcal{F}_0^{0,+}$, by Example 2.34(i) and Theorem 4.14, $P_x$-a.s.

$$\mathbb{1}_A = \mathbb{E}_x[\mathbb{1}_A \mid \mathcal{F}_0^{0,+}] = \mathbb{E}_x[\mathbb{1}_A \mid \mathcal{F}_0^0] = \mathbb{E}_x[\mathbb{E}_x[\mathbb{1}_A \mid \mathcal{F}_0^0]] = P_x(A),$$

so $P_x(A) \in \{0,1\}$. ∎

**Remark 4.17**

A general BM as in Definition 4.1 (started at $x$) also has the Markov property w.r.t. $(\mathcal{F}_t^{0,+})$. Since a shift need not exist, it is formulated as

$$\mathbb{E}_x\Big[\prod_{i=1}^n \mathbb{1}_{A_i}(B_{t_i+s}) \,\Big|\, \mathcal{F}_s^{0,+}\Big] = \mathbb{E}_{B_s}\Big[\prod_{i=1}^n \mathbb{1}_{A_i}(B_{t_i})\Big] \qquad P_x\text{-a.s.}, \tag{4.19}$$

for cylinder sets. The proof of (4.16) used only (4.9) and a.s. continuity, which any BM satisfies (Definition 4.1(i)–(iii)). In particular $\mathbb{E}_x[f(B_{t+s}) \mid \mathcal{F}_s^{0,+}] = \mathbb{E}_{B_s}[f(B_t)]$ for $f \in \mathcal{B}(\mathbb{R})_b$.

## 4.4 The $d$-dimensional Brownian motion

**Motivation.** $d$ independent coordinates; everything carries over.

Let $\|\cdot\|$ be the Euclidean norm and

$$p_t(x,y) := \frac{1}{(\sqrt{2\pi t})^d} e^{-\frac{\|y-x\|^2}{2t}} \quad (x,y \in \mathbb{R}^d,\ t>0), \qquad p_0(x,y)\,dy := \delta_x(dy). \tag{4.20}$$

Section 4.2 goes through with this kernel (Kolmogorov extension for Polish-valued processes; Le Gall, Thm. 6.3). Equivalently, if $P_0$ is the one-dimensional Wiener measure on $(C(\mathbb{R}_+), \mathcal{B}(C(\mathbb{R}_+)))$ and $P^{(i)} = P_0$, $i = 1,\dots,d$, then $\mathbf{P}_0 := \bigotimes_{i=1}^d P^{(i)}$ is the ***d*-dimensional Wiener measure** on $\Omega := C(\mathbb{R}_+)^d = C(\mathbb{R}_+, \mathbb{R}^d)$, $\mathcal{F} := \mathcal{B}(C(\mathbb{R}_+)^d)$. Under $\mathbf{P}_0$, $W_t(\omega) := \omega(t) = (\omega_1(t), \dots, \omega_d(t))$ is the canonical $d$-dimensional BM from 0, and under $P_x := \mathbf{P}_0 \circ \Pi_x^{-1}$, $\Pi_x(\omega) = \omega + x$, it starts at $x$. With $\mathcal{F}_t^0 := \sigma(W_s \mid s \le t)$ and $\mathcal{F}_t^{0,+} := \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^0$ (again filtrations to which $W$ is adapted),

$$P_x((W_{t_1}, \dots, W_{t_n}) \in A_1 \times \cdots \times A_n) = \int_{A_1} \cdots \int_{A_n} \prod_{i=1}^n p_{t_i - t_{i-1}}(x_{i-1}, x_i)\, dx_1 \cdots dx_n \tag{4.21}$$

for $0 \le t_1 < \cdots < t_n$, $A_i \in \mathcal{B}(\mathbb{R}^d)$, $x_0 = x$, $t_0 = 0$, with $d$-dimensional Lebesgue measure $dx_i$. (4.14), (4.15), Theorem 4.14, Remark 4.15, Theorem 4.16 carry over. The $W^i$ are independent BMs: (4.21) factorises, so $P_x(\bigcap_{i=1}^d \tilde{A}_i) = \prod_i P_x(\tilde{A}_i)$ for cylinder sets $\tilde{A}_i = \{(W_{t_1}^i, \dots, W_{t_n}^i) \in B_1^i \times \cdots \times B_n^i\}$, and Theorem 3.10 extends this to the generated $\sigma$-algebras.

**Definition 4.18 ($d$-dimensional Brownian motion)**

An $\mathbb{R}^d$-valued process $(W_t)_{t\geq 0}$ on some $(\Omega, \mathcal{F}, P_x)$ is a **$d$-dimensional BM started at $x$** if (i) $P_x(W_0 = x) = 1$; (ii) for $0 \leq s < t$, $W_t - W_s \sim N(0, (t-s)I_d)$, i.e. $P_x(W_t - W_s \in A) = \int_A p_{t-s}(0,y)\,dy$; (iii) the increments $W_{t_1} - W_0, \ldots, W_{t_n} - W_{t_{n-1}}$ are independent for $0 \leq t_1 < \cdots < t_n$; (iv) almost all paths are continuous.

**Remark 4.19**

A $d$-dimensional BM satisfies (4.21) ( $\Longleftrightarrow$ Definition 4.18(i)–(iii)), so has the canonical finite-dimensional laws, with only $P_x$-a.e. continuous paths. As in Remark 4.17 it has the Markov property

$$\mathbb{E}_x\Big[\prod_{i=1}^n \mathbb{1}_{A_i}(W_{t_i+s}) \,\Big|\, \mathcal{F}_s^{0,+}\Big] = \mathbb{E}_{W_s}\Big[\prod_{i=1}^n \mathbb{1}_{A_i}(W_{t_i})\Big] \quad P_x\text{-a.s.}, \tag{4.22}$$

and in particular $\mathbb{E}_x[f(W_{t+s}) \mid \mathcal{F}_s^{0,+}] = \mathbb{E}_{W_s}[f(W_t)]$ for $f \in \mathcal{B}(\mathbb{R}^d)_b$.